

Project Design Phase

Proposed Solution Template

- **Date:** 18 June 2026
- **Project Name:** Shopez

Proposed Solution Template

The project team has filled out the proposed solution parameters for Shopez below:

S.No.	Parameter	Description
1	Problem Statement	E-commerce websites are often slow, hard to manage, insecure due to exposed credentials, and fragile when back-end databases crash.
2	Idea / Solution description	Shopez: A lightweight MERN stack e-commerce web app featuring a secure role-based login, interactive animated credit card payment checkout, robust local JSON database fallback, and clean dashboard insights.
3	Novelty / Uniqueness	The integration of a seamless automatic JSON fallback system when MongoDB connectivity fails, ensuring high reliability, paired with modern micro-animations in the payment UI.
4	Social Impact / Customer Satisfaction	Enhanced shopper trust due to secure credential policies, readable layout, and interactive, error-resistant checkout flow.
5	Business Model	Transaction fees on customer payments and monthly subscription tier plans for storefront admins.
6	Scalability of the Solution	Easily deployable via Docker containers. Client React UI is hosted on static CDN while Node server runs independently.